

CHARLES DAIL

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Principal Software Engineer

Principal Software Engineer and hands-on technical leader with 25+ years architecting and delivering EHR systems, clinical workflows, HIPAA-compliant mobile apps, secure healthcare APIs, interoperability solutions (HL7, FHIR familiarity), and provider workflows across mobile, web, and cloud platforms.

Work Experience

Veradigm

Oct 2016 - Jun 2026

Principal Software Engineer, Raleigh, NC

- Defined and executed the company's first mobile strategy, reducing a multi-million-dollar sales risk and aligning cross-team priorities across architecture, product, and clinical stakeholders.
- Partnered with architecture, product, and clinical analysts to define MVP acceptance criteria and sequence 5+ early-release clinical workflows, accelerating MVP readiness.
- Architected and delivered Xamarin-based iOS and Android apps, implementing a suite of core clinical workflows (secure messaging, ePrescribing/EPCS, chart review, documentation, imaging) and establishing a cross-platform mobile strategy adopted across the EHR product line. Cited in 10+ sales wins and contributing to increased EHR suite adoption.
- Co-architected and implemented a HIPAA-compliant EHR Web API with 100+ secure endpoints, enabling unified access for desktop, mobile, and web clients and reducing integration time for new features.
- Architected an Azure Relay proxy bridging enterprise on-prem clinical environments to cloud/mobile platforms, improving secure PHI data flow and reducing manual integration effort by 40%.
- Ported high-value EHR workflows into the browser-based EHR (Node.js/Ember.js), delivering prioritized functionality within 8 weeks and enabling rapid sales demonstrations.
- Standardized core GitHub repositories and CI/CD pipelines, reducing build failures by 25% and cutting deployment friction across multiple engineering teams.
- Integrated automated code-quality checks, security scans, and staged environments, reducing production defects by 20% and improving issue visibility across the development lifecycle.
- Implemented development containers, reducing onboarding time from hours to minutes and eliminating environment-related issues for 8+ engineers.
- Mentored 8+ engineers through structured onboarding and regular code reviews, reducing ramp-up time by 50% and improving code quality across mobile and web teams.
- Led two Scrum teams through 100+ sprints, maintaining an average 25-point velocity and delivering sprint-end demos that improved stakeholder alignment and reduced rework.
- Leveraged AI code-generation tools and enforced PR documentation standards, reducing review cycles by 40% and accelerating integration across mobile and web teams.

Veradigm

May 2014 - Oct 2016

Expert Software Engineer, Raleigh, NC

- Built a proof-of-concept iOS app, JWT-secured web API, and cloud proxy demonstrating cross-platform clinical messaging, validating a new mobile opportunity within 1 week.
 - Revitalized a stalled initiative by resetting technical direction and delivering a WPF solution with COM wrappers, reducing integration effort with legacy Delphi systems and improving provider workflows by 30%.
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Skills & Interests

Programming Languages: C#, JavaScript, TypeScript, SQL, Pascal (Delphi)

Backend & Architecture: Microservices, Apache Kafka, Debezium, Transactional Outbox Pattern, Node.js, Express, ASP.NET, Web API, RESTful API Design, Orchestration Patterns, JWT authentication, OAuth2, EHR Web API

Frontend Development: Ember.js, HTML5, CSS, WinForms, WPF

Cloud & DevOps: Azure, Azure App Services, Azure Key Vault, Azure Relay, Docker, Development Containers, GitHub, Azure DevOps Server, CI/CD Pipelines

Tools & Infrastructure: Jira, Confluence, Proxmox VE

Methodologies & Software Practice: Test-Driven Development (TDD), Agile/Scrum

Operating Systems: macOS, Linux, Windows

Leadership: Servant Leadership, Mentoring & Coaching, Cross-team Collaboration, Team resilience

Healthcare Technologies: CCD/CDA, EPCS, HIPAA compliance, PHI protection, clinical workflows, EHR systems, audit logging, secure clinical data exchange, HL7/FHIR (familiarity), Clinical decision support (CDS), Azure Key Vault (PHI security)

Healthcare Domain Expertise

- EHR systems (desktop, web, mobile)
- Clinical workflows: messaging, prescribing/EPCS, chart review, documentation, imaging.
- HIPAA compliance & PHI protection
- Healthcare interoperability standards (HL7, FHIR) — familiarity with data exchange formats and workflows
- Secure clinical data exchange
- Provider workflows & care coordination
- Mobile healthcare applications

Education

Wake Technical Community College, Raleigh, NC, Associate of Science in Computer Programming, May 1997

Volunteering

Building homes, Habitat for Humanity, Raleigh, NC
Assisted with building homes for low income families.

Food drives, local food banks, Raleigh, NC
Packaged and delivered meals for people in need.

Other

Conferences & Workshops

- Big Nerd Ranch conference focusing on iOS development and team building.
- Continuous learning via Pluralsight, LinkedIn Learning, Udemy and in-person networking events.